P1 DAY 6 - EXAM PRACTICE 1

# Structured practice paper 1

Foundation and secure method - 50 marks - 75 minutes

YOUR AIM TODAY

## Complete a realistic structured paper on quadratics, functions, coordinate geometry and circles.

YOUR RULE

Closed book. Calculator allowed. Answer all eight questions. Show essential working and stop after 75 minutes. Keep the solution section closed.

Student: Rohan

Start time: ____________ Finish time: ____________

Date: __________________

INSTRUCTIONS

# Candidate information

This paper uses linked parts and explicit mark allocations like an examination.

| | |
|---|---|
| Time | 75 minutes |
| Marks | 50 |
| Questions | 8 compulsory structured questions |
| Method | Show algebra and exact values; unsupported answers may lose marks |
| Checking | Reserve the final 7 minutes |

| Question | Topic | Marks |
|---|---|---|
| 1 | Quadratic forms | 7 |
| 2 | Discriminant and repeated roots | 6 |
| 3 | Quadratic inequalities | 6 |
| 4 | Composite functions | 7 |
| 5 | Inverse function | 6 |
| 6 | Coordinate geometry | 6 |
| 7 | Parallel and perpendicular lines | 6 |
| 8 | Circle and tangent | 6 |
| | Total | 50 |

**EXAM RULE**

If stuck, write the relevant formula or structure, leave space and continue. Return before time ends.

QUESTIONS 1-2

# Structured questions

Running total after this page: 13 marks.

QUESTION 1

## Quadratic forms

[7 marks]

**(a) Expand and simplify $(2x-1)(x+5)$. [2]**

**(b) Complete the square: $x^2-4x-5$. [2]**

**(c) State the turning point of $y=x^2-4x-5$. [1]**

**(d) Hence solve $x^2-4x-5=0$. [2]**

QUESTION 2

## Discriminant and repeated roots

[6 marks]

**(a) Find the discriminant of $2x^2+3x+5=0$ and state the number of real roots. [3]**

**(b) Find k if $x^2-8x+k=0$ has a repeated root. [3]**

# Structured questions

Running total after this page: 26 marks.

QUESTION 3

## Quadratic inequalities

[6 marks]

**(a) Solve $x^2-7x+6\le 0$. [3]**

**(b) Solve $2x^2+7x+3>0$. [3]**

QUESTION 4

## Composite functions

[7 marks]

GIVEN

$f(x)=2x+3$ and $g(x)=x^2-1$.

**(a) Find $f(-4)$. [1]**

**(b) Find $fg(x)$. [2]**

**(c) Find $gf(x)$. [2]**

**(d) Show clearly that $fg(x)$ is not equal to $gf(x)$. [2]**

# Structured questions

Running total after this page: 38 marks.

QUESTION 5

## Inverse function

[6 marks]

GIVEN

f(x)=5-2x.

**(a) Find f^-1(x). [3]**

**(b) Find f^-1(9). [1]**

**(c) Verify that f(f^-1(x))=x. [2]**

QUESTION 6

## Coordinate geometry

[6 marks]

GIVEN

A=(1,-2) and B=(7,6).

**(a) Find the gradient of AB. [2]**

**(b) Find the midpoint of AB. [1]**

**(c) Find the exact length AB. [2]**

**(d) Write an equation of AB. [1]**

QUESTIONS 7-8

# Structured questions

Running total after this page: 50 marks.

QUESTION 7

## Parallel and perpendicular lines

[6 marks]

GIVEN

Use A and B from Question 6.

**(a) Find the perpendicular bisector of AB. [3]**

**(b) Find the line through (-2,5) parallel to AB. [3]**

QUESTION 8

## Circle and tangent

[6 marks]

GIVEN

C: $(x-1)^2+(y+2)^2=36$.

**(a) State the centre and radius. [2]**

**(b) Show that P=(1,4) lies on C. [2]**

**(c) Find the tangent to C at P. [2]**

STOP

# Solution section locked

Do not continue until the full paper has an attempt and 75 minutes have ended.

**BEFORE OPENING**

Write your predicted score and identify the two questions you found hardest.

Predicted score: ____ / 50. Hardest questions:

**MARKING METHOD**

Use another colour. Circle the first wrong line. After reading the explanation, close the booklet and redo the entire part.

# Easy worked explanations

Equivalent correct methods are acceptable.

## Question 1 - Quadratic forms [7 marks]

| Part | Easy working and answer |
| --- | --- |
| a | Four products: $2x^2+10x-x-5=2x^2+9x-5$. |
| b | Half of -4 is -2: $x^2-4x-5=(x-2)^2-4-5=(x-2)^2-9$. |
| c | The square is zero at $x=2$, so the turning point is $(2,-9)$. |
| d | Set $(x-2)^2-9=0$. Then $(x-2)^2=9$, so $x-2=\pm 3$ and $x=5$ or $-1$. |

## Question 2 - Discriminant and repeated roots [6 marks]

| Part | Easy working and answer |
| --- | --- |
| a | $a=2,b=3,c=5$. $D=3^2-4(2)(5)=9-40=-31$. Because $D<0$, there are no real roots. |
| b | Repeated root means $D=0$. $D=(-8)^2-4k=64-4k$. Set $64-4k=0$, so $k=16$. |

SOLUTIONS 3-4

# Easy worked explanations

Equivalent correct methods are acceptable.

## Question 3 - Quadratic inequalities [6 marks]

| Part | Easy working and answer |
|---|---|
| a | Factor: $(x-1)(x-6)<=0$. The upward parabola is non-positive between the roots, including them: $1<=x<=6$. |
| b | Factor: $(2x+1)(x+3)>0$. Roots are -3 and -1/2. The upward parabola is positive outside: $x<-3$ or $x>-1/2$. |

## Question 4 - Composite functions [7 marks]

| Part | Easy working and answer |
|---|---|
| a | $f(-4)=2(-4)+3=-8+3=-5$. |
| b | Apply g first: $fg(x)=f(x^2-1)=2(x^2-1)+3=2x^2+1$. |
| c | Apply f first: $gf(x)=g(2x+3)=(2x+3)^2-1=4x^2+12x+8$. |
| d | The simplified expressions $2x^2+1$ and $4x^2+12x+8$ are different, so the composites are not equal. |

# Easy worked explanations

Equivalent correct methods are acceptable.

## Question 5 - Inverse function [6 marks]

| Part | Easy working and answer |
|---|---|
| a | Write y=5-2x. Swap x and y: x=5-2y. Then 2y=5-x, so f^-1(x)=(5-x)/2. |
| b | f^-1(9)=(5-9)/2=-4/2=-2. |
| c | f(f^-1(x))=5-2[(5-x)/2]=5-(5-x)=x. This confirms the functions undo each other. |

## Question 6 - Coordinate geometry [6 marks]

| Part | Easy working and answer |
|---|---|
| a | m=(6-(-2))/(7-1)=8/6=4/3. |
| b | Midpoint=((1+7)/2,(-2+6)/2)=(4,2). |
| c | Changes are 6 and 8, so AB=sqrt(6^2+8^2)=10. |
| d | Using A: y+2=(4/3)(x-1). Any equivalent equation is correct. |

# Easy worked explanations

Equivalent correct methods are acceptable.

## Question 7 - Parallel and perpendicular lines [6 marks]

| Part | Easy working and answer |
| --- | --- |
| a | AB has m=4/3, so perpendicular m=-3/4. It passes through midpoint (4,2): y-2=(-3/4)(x-4). |
| b | A parallel line has m=4/3. Through (-2,5): y-5=(4/3)(x+2). |

## Question 8 - Circle and tangent [6 marks]

| Part | Easy working and answer |
| --- | --- |
| a | Centre (1,-2), radius sqrt(36)=6. |
| b | At P: (1-1)^2+(4+2)^2=0+36=36, so P lies on C. |
| c | The radius from (1,-2) to (1,4) is vertical, x=1. The tangent is horizontal through P, so y=4. |

CORRECTION RECORD

# Turn marks into the next lesson

Do not record only the final answer.

| Q/part | First wrong line | Error type | Correct rule |
| --- | --- | --- | --- |
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| Topic | Score | Available | % |
| --- | --- | --- | --- |
| Quadratics | | 19 | |
| Functions | | 13 | |
| Coordinate geometry | | 12 | |
| Circles | | 6 | |
| Total | | 50 | |

**RETEST**

Redo every lost mark tomorrow without looking. Any topic below 60% returns to the notes booklet first.